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Smart Campus Event Hub: AI-Based Recommendations with Timetable-Aware Notifications

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Abstract

“The Smart Campus Event Hub is designed to enhance student involvement in campus activities through intelligent event management and user-specific recommendations.” The system combines user profiles, event data, and timetable analysis to suggest events that align with both individual interests and available time slots. By considering schedule constraints, the platform minimizes conflicts and e

nsures that recommended events are practically feasible for students to attend. An integrated recommendation module analyzes user preferences and past interactions to deliver relevant suggestions, while an automated notification system provides timely alerts through both in-app messages and email. These notifications are tailored to each user's free time, increasing the likelihood of participation. The platform is developed using Flask, SQLAlchemy, and supporting AI components, ensuring efficient performance and scalability. Evaluation of the system indicates improved event discovery, better utilization of free time, and higher user engagement compared to traditional event management approaches. The proposed solution demonstrates how combining personalization with schedule awareness can significantly enhance the overall campus experience.

Introduction

Educational institutions organize various academic, technical, and cultural events that enhance student learning beyond the classroom. However, students often struggle to find events that match their interests and fit their schedules. Existing event management systems mainly display generic event lists, requiring manual searching and filtering, which is time-consuming and often leads to missed opportunities. These systems also lack integration with student timetables, resulting in scheduling conflicts and ineffective, non-personalized notifications.

To overcome these limitations, the **Smart Campus Event Hub** is proposed. The system combines personalization and schedule awareness by analyzing user interests, past interactions, and academic timetables. It recommends events that are both relevant and feasible to attend by identifying free time slots and matching them with suitable events. The platform also provides personalized email and in-app notifications, ensuring students receive timely and meaningful updates.

The proposed system uses a content-based recommendation approach, timetable analysis, and gamification features such as points, badges, and leaderboards to encourage participation. Its architecture consists of a user-friendly frontend, a Flask-based backend, a SQLAlchemy database, and AI-powered recommendation and scheduling modules.

Performance evaluation shows that the system operates efficiently, with response times ranging from approximately **20 ms for page loading** to **200 ms for recommendation generation**. Results indicate improved event discovery, reduced scheduling conflicts, enhanced user engagement, and higher p

articulation rates compared to traditional campus event management systems. Overall, the Smart Campus Event Hub provides a more intelligent, user-centered, and effective solution for campus event management.

Conclusion

The Smart Campus Event Hub successfully addresses the limitations of traditional event management systems by integrating artificial intelligence, timetable analysis, and personalized notifications. The system enhances event participation, improves time management, and provides a user-centric experience. By aligning event recommendations with user interests and availability, the platform ensures higher relevance and usability. Future work can focus on incorporating advanced machine learning models, real-time analytics, and mobile application support to further enhance system capabilities. The proposed solution demonstrates the potential of intelligent systems in transforming campus event management into a more efficient and engaging process.

References

- [1] N. M. Bawankar, A. T. Tembhurne, G. R. Chatap, M. M. Bandbuche, T. S. Dhote, and P. R. Surjuse, "EVENT MANAGEMENT SYSTEM," 2021. [Online]. Available: www.ijariie.com128
- [2] M. Q. Joseph Mayan-ao et al., "Development of Events Management System." [Online]. Available: <https://www.ijert.org/>
- [3] K. Acharya, "AN INTERNSHIP REPORT ON EVENT MANAGEMENT SYSTEM PROJECT." [4] V. Mishra et al., "Event Management System." [Online]. Available: www.ijtrd.com
- [5] B. Gupta and M. P. Vani, "An Overview of Web Sockets: The future of Real-Time Communication," International Research Journal of Engineering and Technology, vol. 434, 2008, [Online]. Available: www.irjet.net
- [6] P. Mahindrakar* and Dr. U. Pujeri, "Insights of JSON Web Token," International Journal of Recent Technology and Engineering (IJRTE), vol. 8, no. 6, pp. 1707–1710, Mar. 2020, doi: 10.35940/ijrte.f7689.038620.
- [7] "AI Powered Event Management Platform," International Research Journal of Modernization in Engineering Technology and Science, Feb. 2024, doi: 10.56726/irjmets49378.
- [8] "Barcode Enabled Event Management System for Logistics and Consumables Management - PDF Free Download." Accessed: Mar. 07, 2026. [Online]. Available: <https://docplayer.net/9354095-Barcodeenabled-event-management-system-for-logistics-andconsumables-management.html>
- [9] S. V Sonekar, P. Rohan Kokate, M. Rohini, and I. Uikey, "Design and Implementation of a Web- Based Campus Event Management System," 2025. [Online]. Available: www.ijnrd.org
- [10] M. B. Piedade and M. Y. Santos, "St

udent Relationship Management: Concept, practice and technological support,” IEMC-Europe 2008 -
(<https://www.ijraset.com/>)
2026 International Engineering Management Conference, Europe: Managing Engineering, Tec
hnology and Innovation for Growth, 2008, doi: 10.1109/IEMCE.2008.4618026.

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